Episodic Memory for Robots in the Wild: Benchmarking and Evaluation Across Unstructured Environments

Vedansh Malhan, *Student, Monash University*, William Ngo, *Student, Monash University*

***Abstract*—Vision-Language Models (VLMs) are seeing increased adoption in modern systems such as autonomous vehicles and embodied agents. These systems require a continuous, long-term understanding of past events, known as episodic memory, to maintain task context and prevent operational failures. Despite its critical importance, current benchmarks evaluating episodic memory in VLMs remain limited. Existing datasets typically feature short-duration clips (seconds to minutes) in event-dense environments, characterised by high-frequency occurrences that result in high Signal-to-Noise Ratio (SNR) scenarios. Consequently, there is a significant lack of benchmarks addressing low-SNR, event-sparse environments such as forests, rural roads, and tunnels which demand fine-grained episodic discrimination and long-range temporal retrieval over extended sequences. To address this gap, we propose a benchmarking framework for evaluating episodic memory in modern VLMs in low-SNR environments, utilising extended duration videos ranging from 6 minutes to 2 hours. The framework includes a pipeline for generating natural language query-based question-answer pairs using modern Large Language Models (LLMs) and VLMs, followed by human verification. The benchmark targets three key evaluation categories: sparse-event localisation, temporal ordering, and object counting. This benchmark is anticipated to provide an evaluation framework capable of exposing the limitations of current multi-modal architectures. Preliminary projections suggest that modern VLMs will face significant challenges, with expected maximum recall ($R@1$) scores of 70% for sparse event localisation [1], 80% for temporal reasoning [2] [1] [3], and 50% for object counting [4] [5].**

***Index Terms*—Enter key words or phrases in alphabetical order, separated by commas. For a list of suggested keywords, send a blank e-mail to keywords@ieee.org or visit http://www.ieee.org/organizations/pubs/ani_prod/keywrd98.txt**

I. Introduction

RECENT advancements in Vision-Language Models (VLMs) have facilitated their integration into complex, real-world applications, particularly within the autonomous vehicle and robotics industries [6]. For these systems to operate reliably, the capacity to retain and retrieve specific past experiences over extended periods is crucial. This is known as episodic memory, originating from cognitive psychology [7], episodic memory is defined as the ability to encode, store, and consciously recall specific past events anchored in their precise spatial and temporal contexts. For instance, in autonomous navigation, the inability to recall past road conditions in a GNSS-denied environment poses critical safety risks and compromises localisation capabilities. Similarly, episodic memory in robotics enables long-term continuity, enhanced social interaction, and advanced cognitive planning by allowing agents to store and retrieve specific temporal experiences, rather than relying solely on generalised semantic memory.

Although recent research has benchmarked the temporal and memory capabilities of VLMs, existing datasets, such as Ego4D [3], predominantly focus on short-duration video clips averaging approximately eight minutes. Furthermore, these datasets are heavily skewed toward event-dense settings characterised by a high Signal-to-Noise Ratio (SNR). In the context of video analysis, SNR refers to the density of meaningful, distinct events (the "signal") relative to continuous, repetitive, or uninformative background frames (the "noise"). Consequently, current benchmarks primarily capture general daily activities in object-rich environments like private homes and offices, where visual stimuli and new events occur almost continuously.

Consequently, a significant gap exists regarding the evaluation of VLMs in low SNR, lengthy-context, and event-sparse environments. Scenarios such as navigating through forests, rural roads, and tunnels present a unique challenge. The absence of distinctive landmarks deprives VLMs of visual anchors; as a result, the context window becomes saturated with highly similar tokens. This homogeneity severely challenges frame-level recall and temporal anchoring. Fine-grained episodic discrimination over extended sequences remains largely untested in current VLM literature.

To address this critical gap, this study is guided by the following research question: ***Can modern Vision-Language Models accurately recall, distinguish, and reason with episodic memories in an event-sparse environment?***

The primary objective of this project is to develop a comprehensive and reliable benchmark to assess the episodic memory of VLMs within the aforementioned domains. To achieve this, an automated pipeline is proposed to generate question-answer pairs based on long-form video inputs, which are subsequently validated via human annotation. The benchmark evaluates models across three main categories:

- **Sparse Event Localisation:** Verifying the occurrence

Special thanks to Kavi Katuwandeniya and Dana Kulic for their support in the research. This work was supported in part by Monash University Engineering Department

The next few paragraphs should contain the authors' current affiliations, including current address and e-mail. For example, F. A. Author is with the National Institute of Standards and Technology, Boulder, CO 80305 USA (e-mail: author@boulder.nist.gov).

V. M. Author is with the Software Engineering Department, Monansh University, Clayton Melbourne, Vic 3169 Australia (e-mail: vedanshmalhan03@gmail.com).

of rare, outlier events over extended durations (e.g., identifying a wildlife crossing during a long drive).
- **Temporal Order:** Assessing the model's capacity to recall the chronological sequence of past events.
- **Object Counting:** Testing the model's ability to track and uniquely count distinct objects over time.

By exposing vulnerabilities in current architectures, this benchmark aims to push the boundaries of long-term temporal context retention in modern multi-modal models.

II. Related Work

THE capabilities of Vision Language Models (VLMs) and their temporal reasoning abilities have seen a massive growth in recent years through the development of transformers. However, literature predominantly focuses on short-duration, high-density event scenarios, leaving a critical gap in long-term, low-SNR episodic memory evaluation.

Early video question answering benchmarks such as such as **TGIF** [9], **Movie-QA** [14], and **ActivityNet-QA** [10] laid the groundwork for temporal reasoning in VLM models. Subsequent efforts expanded into instructional and narrated content, with datasets like **HowTo100M** [15] and **CrossTask** [16] evaluating the alignment of spoken language with visual steps. However, these datasets primarily consist of brief clips, typically ranging from a few seconds to five minutes, which are insufficient for testing long-term memory retention. However, these datasets primarily consists of very short clips, often ranging around 10 seconds to 5 minutes.

More recently, the focus in benchmarks have shifted more towards comprehensive and long-context video evaluations. Benchmarks such as **Video-MME** [13], **LongVideoBench** [17], and **EgoSchema** [12] provide extensive multi-modal evaluations across longer durations. While these benchmarks test temporal contexts, they are fundamentally built upon high-SNR environments (e.g., movies, vlogs, and television shows) where distinct visual and audio cues occur with high frequency. Furthermore, these datasets are primarily sourced from third-person media, limiting their applicability to autonomous vehicles and robotics, which inherently rely on egocentric viewpoints to perform real-world tasks.

In the context of egocentric vision and embodied AI's, which are more relevant to robotics. The current frontier benchmarks includes **Ego4D** [3], incorporating test for egocentric video understanding using body mounted cameras from volunteers. In simulated environments, benchmarks such as **Embodied Question Answering** [18], **House3D** [19], and **Enter the Mind Palace** [20] all evaluate spatial and episodic memory for embodied agents. While these benchmarks are highly influential, these datasets generally remain focused on human activities that are high SNR and bounded simulations. Similarly, in the autonomous driving domain, **NuScenes QA** [21] and **The blind Spot of Adaptation** [22] benchmarks episodic memory in autonomous vehicles, but strictly focuses on dense, high SNR urban traffic, rather than feature sparse environments.

Despite these new benchmarks in various domains of advancements, a critical gap remains. None of these benchmarks evaluate long term temporal context in unbounded, low SNR and egocentric environments, which is the primary focus of this benchmark.

III. Methodology

IV. Pipeline Architecture

We present a benchmark for assessing the episodic memory of Vision-Language Models (VLMs) in long, egocentric dash-cam recordings. Each recording is a forward-facing vehicle camera sequence of 1 to 6 minutes, captured in one of three environments: forests and rural roads. These environments have low event density and few stable visual landmarks. A model answering questions about such a recording cannot rely on a single frame and must recall what occurred across a long time span. The benchmark targets three categories:

1) **Sparse event localisation**, which tests whether a model can verify rare, outlier events without hallucinating them.
2) **Temporal order**, which tests whether a model recalls the sequence of events rather than inferring it from priors.
3) **Counting**, which tests whether a model can track the number of distinct objects it has observed across the whole recording.

At the time of writing, the sparse-event and temporal-order categories are implemented. Counting category had run into limitation which we will later discuss.

A. Data Source and Preparation

Source frames are taken from the Boreas dataset **[TBD: citation]** and a forest sequence (K-01) **[TBD: citation or source]**. Each frame is a timestamped image whose filename encodes its capture time in epoch microseconds, for example `1733343593917869.png`. Elapsed time is computed from these filenames rather than estimated from the content of the video.

Frames are partitioned into sections with a sliding window. Within a section, `frames_per_section` frames are sampled at stride `step`. The next section begins `skip` frames after the last sampled frame of the previous section. For the sparse configuration, we use `frames_per_section` $= 7$, `step` $= 8$, and `skip` $= 2$. For the temporal configuration, we use `frames_per_section` $= 5$, `step` $= 10$, and `skip` $= 10$. Each section's start and end times in seconds are recorded in a `sections.json` manifest. Sampled frames are re-encoded as JPEG at quality 80 without resizing, so that the original resolution is preserved. This is done to remain within budget and model context limit contraints.

B. Ground-Truth Construction

Ground truth is built in four stages. Each stage is run separately, and a human reviewer checks the output before the next stage.

Benchmarks	#Videos	#Clips	Len.(s)	#QA Pairs	Anno.
MSRVTT-QA [8]	2,990	2,990	15.2	72,821	A
TGIF-QA [9]	9,575	9,575	3.0	8,506	A&M
ActivityNet-QA [10]	800	800	111.4	8,000	M
MVBench [11]	3,641	3,641	16.0	4,000	A
EgoSchema [12]	5,063	5,063	180.0	5,063	A&M
Video-MME-S	300	300	80.7	900	M
Video-MME-M	300	300	515.9	900	
Video-MME-L	300	300	2466.7	900	
Video-MME [13]	900	900	1017.9	2,700	
FYP2026S1-3073	**132**	**132**	**2400**	**TBD**	**A&M**

TABLE I

THIS COMPARISON TABLE SHOWS SEVERAL KEY ASPECTS THAT OTHER BENCHMARKS HAVE TESTED AGAINST OURS: THE TOTAL NUMBER OF VIDEOS **(#VIDEOS)**, THE AVERAGE LENGTH OF VIDEO INPUT **(LEN.(S)**, THE TOTAL NUMBER OF CLIPS **(#CLIPS)**, THE TOTAL NUMBER OF QUESTION-ANSWER PAIRS AVAILABLE FOR EVALUATION **(#QA PAIRS)**, METHOD OF ANNOTATION (M FOR MANUAL AND A FOR AUTOMATED) **(# ANNO.)**

1) Candidate Extraction: A VLM is given each section's frames together with a task-specific prompt and returns structured JSON. In the sparse task, the prompt asks for rare or noteworthy events and instructs the model to ignore normal driving. In the temporal task, the prompt asks for a section summary and a chronological list of timestamped events. Each extracted event is assigned `start_seconds` and `end_seconds` from its section's manifest entry. The model is not asked to produce timestamps.

2) Deterministic Ordering and LLM-Assisted Merging: Detections are sorted in Python by the timestamp embedded in their frame filename. The sorted list is then output into a JSON `all_results` this is then passed to an LLM in windows of `review_window` (sparse) or `merge_window` (temporal) items. In the sparse task, the LLM merges duplicate observations of the same event across section boundaries and removes spurious detections, each with a recorded reason. In the temporal task, the LLM only merges continuing or duplicate observations. The prompts forbid reordering and forbid inventing events. Seconds are reattached from the original records after merging, so merged records retain their original time references. The output is stored as a merged list (`events.json` or `timeline.json`).

3) Human Verification: A reviewer checks each merged list (`events.json` or `timeline.json`) against the referenced frames and edits or removes items as needed. Only verified items may proceed to question generation. For the temporal task, a prose storyline is also produced to support this review.

4) Question Generation: The LLM generates questions only from the verified list. Each question includes its type, the event IDs and frame evidence on which it is based, its answer options, and the indices of the correct options.

C. Question Design

Sparse event localisation. Three question types are generated:

- *Existence:* asks whether a verified event occurred during the drive. The answer is yes.
- *Deceptive:* asks whether a plausible but invented event occurred. The answer is always no. Each invented event is used only once, so that no question depends on a reused distractor.
- *Noteworthy:* presents four candidate events. Exactly one is a verified event and three are invented. The answer is the verified event.

Every invented event is recorded in a separate `false_events` array with its `question_id` and a rationale, so that a reviewer can confirm it never occurred.

Temporal order. Three question types are generated:

- *Pairwise order:* asks which of two events occurred first.
- *Sequence order:* asks the model to order three to five events. Grading is order-sensitive.
- *Before/after:* asks which event immediately preceded or followed a given event.

The complete, integrated architecture of our benchmarking framework is illustrated in Fig. 1. This unified layout demonstrates the shared multi-threaded video pre-processing workflow, which seamlessly splits long-form data before parallelising downstream evaluations into our three distinct task-specific domains (Sparse Event, Temporal, and Object Counting).

D. Selection of Evaluation Categories

The overall design of the benchmark is based on the cognitive requirements of autonomous agents operating in real world, low SNR environments. Through analysis of existing benchmarks in similar domains, we have identified three critical categories in our assessment of episodic memory that aligns with autonomous vehicles and embodied agents.

- **Sparse Event Localisation:** In repetitive environments (e.g., continuous highway driving), the ability to detect rare anomalies is vital for safety and context awareness.
- **Temporal Ordering:** Long-term planning requires an understanding of the chronological sequence of events separated by extended durations, testing the model's ability to maintain a timeline.
- **Object Counting:** Tracking and uniquely identifying objects over time tests the limits of visual-primitive numeracy and object permanence across long horizons.

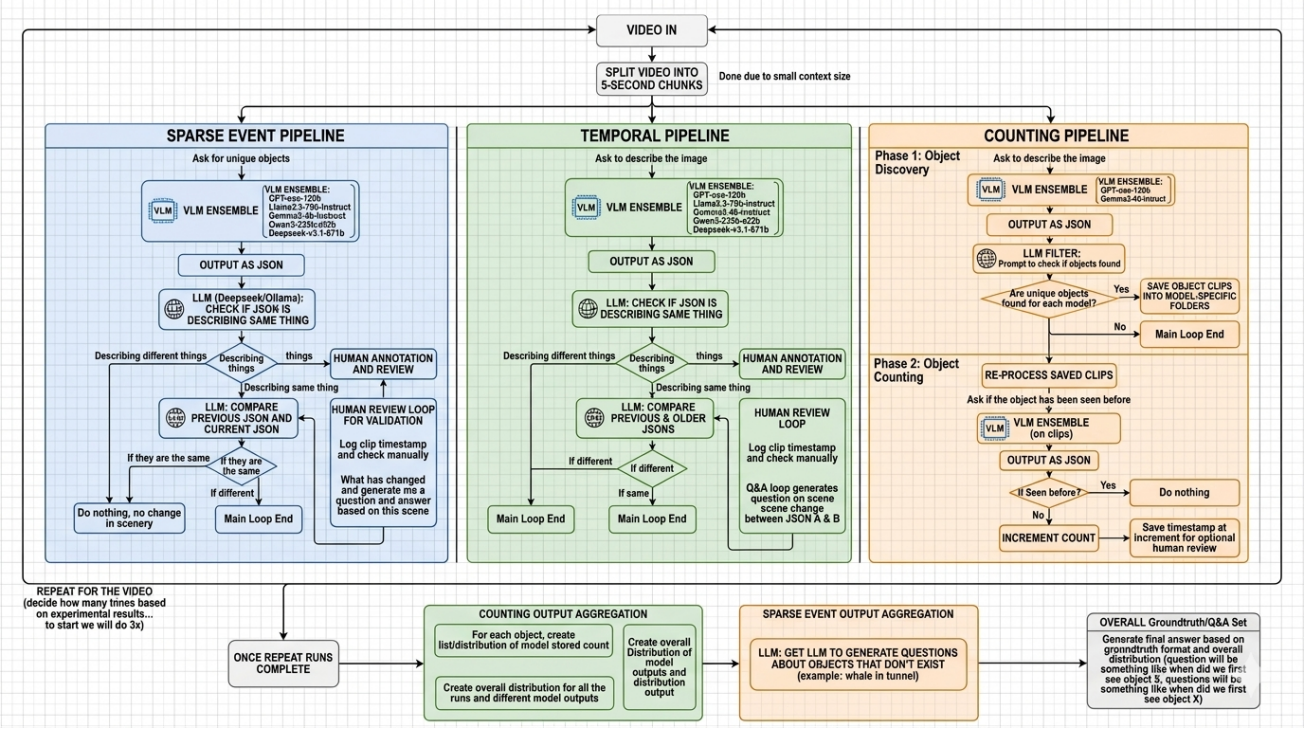


Fig. 1. Comprehensive workflow of the automated benchmarking pipeline, mapping out the shared frontend multi-VLM ensemble stage and subsequent specialised tracking pathways across automated repeat cycles.

V. Initial and Anticipated Results

This benchmark is designed such that models are expected to score as low as possible, thereby revealing gaps in current implementations. We target 60% of evaluated models achieving a maximum $R@1$ score (the first output from the model being the correct one for a given query) below the thresholds specified in Table II. Our projected performance ceilings are empirically grounded in recent diagnostic evaluations of frontier VLM architectures. Specifically, the 50% ceiling for object counting reflects the structural tokenisation limitations highlighted in recent probing frameworks [5], [4], which demonstrate that general-purpose VLMs natively struggle with visual-primitive numeracy. Similarly, our 70% cap on sparse event localisation aligns with severe performance degradation observed in long-horizon egocentric benchmarks [23], [2], [1], where long-context token saturation consistently limits fine-grained temporal anchoring.

TABLE II
Projections of Models' Maximum R@1 Results on the Proposed Benchmark

Category	R@1 Percentage
Counting	50% [5], [4]
Temporal	80% [23], [1]
Sparse Event	70% [2], [23]

References

[1] M. Cai, R. Tan, J. Zhang, B. Zou, K. Zhang, F. Yao, F. Zhu, J. Gu, Y. Zhong, Y. Shang, Y. Dou, J. Park, J. Gao, Y. J. Lee, and J. Yang, "Temporalbench: Towards fine-grained temporal understanding for multimodal video models," *arXiv preprint arXiv:2410.10818*, 2024. [Online]. Available: https://arxiv.org/abs/2410.10818

[2] Anonymous, "Finebench: Evaluating fine-grained temporal anchoring in vlms," *Anticipated/Placeholder Citation*, 2026.

[3] K. Grauman, A. Westbury, E. Byrne *et al.*, "Ego4d: Around the world in 3,000 hours of egocentric video," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[4] M. S. Danish, M. A. Munir, S. R. A. Shah, K. Kuckreja, F. S. Khan, P. Fraccaro, A. Lacoste, and S. Khan, "GEOBench-VLM: Benchmarking vision-language models for geospatial tasks," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. [Online]. Available: https://arxiv.org/abs/2411.19325

[5] Anonymous, "Counting to four: Limitations of tokenization in visual numeracy," *Anticipated/Placeholder Citation*, 2026.

[6] Kodiak Robotics, "Llms take the wheel," *Kodiak.ai News*, 2023. [Online]. Available: https://kodiak.ai/news/llms-take-the-wheel

[7] E. Tulving, "Episodic and semantic memory," in *Organization of memory*, E. Tulving and W. Donaldson, Eds. New York: Academic Press, 1972, pp. 381–403.

[8] J. Xu, T. Mei, T. Yao, and Y. Rui, "Msr-vtt: A large video description dataset for bridging video and language," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[9] Y. Jang, Y. Song, Y. Yu, Y. Kim, and K. Gunhee, "Tgif-qa: Toward spatio-temporal reasoning in visual question answering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[10] Z. Yu, D. Xu, J. Yu, T. Yu, Z. Zhao, Y. Zhuang, and D. Tao, "Activitynet-qa: A dataset for understanding complex web videos via question answering," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019.

[11] K. Li, Y. Wang, Y. He, Y. Li, Y. Wang, Y. Liu, and L. Wang, "Mvbench: A comprehensive multi-modal video understanding benchmark," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[12] K. Mangalam *et al.*, "Egoschema: A diagnostic benchmark for very long-form video language understanding," in *International Conference on Learning Representations (ICLR)*, 2024.

[13] C. Fu, J. Peidong, S.-Y. Wang *et al.*, "Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis," *arXiv preprint arXiv:2405.21075*, 2024.

[14] M. Tapaswi, Y. Zhu, R. Stiefelhagen, A. Torralba, R. Urtasun, and S. Fidler, "Movieqa: Understanding stories in movies through question-answering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
[15] A. Miech, D. Zhukov, J.-B. Alayrac, M. Tapaswi, I. Laptev, and J. Sivic, "Howto100m: Learning a text-video embedding by watching hundred million narrated video clips," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
[16] D. Zhukov, J.-B. Alayrac, R. G. Cinbis, D. Fouhey, I. Laptev, and J. Sivic, "Crosstask: weakly supervised learning from instructional videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
[17] H. Wu *et al.*, "Longvideobench: A benchmark for long-context interleaved video-language understanding," *arXiv preprint arXiv:2407.15743*, 2024.
[18] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra, "Embodied question answering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
[19] Y. Wu, Y. Wu, G. Gkioxari, and Y. Tian, "House3d: A rich and realistic 3d environment," *arXiv preprint arXiv:1801.02209*, 2018.
[20] Anonymous, "Enter the mind palace: Spatial and episodic memory for embodied agents," in *Anticipated/Placeholder Citation*, 2024.
[21] T. Qian *et al.*, "Nuscenes-qa: A multi-modal visual question answering benchmark for autonomous driving scenario," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.
[22] Anonymous, "The blind spot of adaptation: A study on the temporal grounding of vlms," *Anticipated/Placeholder Citation*, 2024.
[23] Z. Wang, Y. Zhang, S. Yu, C. Zhang, Z. Zhao, J. Yoon, H. Lee, G. Bertasius, and M. Bansal, "EgoMemReason: A memory-driven reasoning benchmark for long-horizon egocentric video understanding," 2026. [Online]. Available: https://arxiv.org/abs/2605.09874